

IDS Alert Prioritization and Strengthening

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Introduction

Intrusion Detection Systems monitor network traffic and provide alerts about potentially malicious network traffic. A big problem with these detection systems is preventing false positives. Usually these systems take the approach to make their systems alert too often, because it is easier to deal with false positives rather than accidentally let malware slip through. One potential solution to this would be to use a neural network to classify network traffic, potentially reducing the number of false positives.

Plan

1. Analyze datasets and find correlations between attribute-values.
2. Figure out the best method to train the data
3. Experimenting with different neural network structures and designs.
4. Implement ML methodologies to train further to improve on accuracy
5. Repeat the process several times to enhance the final ML algorithm to be as accurate as possible without overfitting the data

The current plan is to train a neural network to classify packets. Using the neural network to prioritize alerts from a more traditional IDS system.

Issues

Due to the resource intensive nature of training machine learning models from large datasets, powerful hardware is required. We are currently looking into potential solutions.

Schedule

Proposal - February 14
Investigate Datasets - Feb 21
Midterm (progress) report - April 3
Midterm presentation - April 3
Refine Approach - Apr 24
Final deliverables - May 1
Final presentation - TBD
Mentor Meetings - Every Friday

Progress

After learning a bit of the domain knowledge of our dataset, we first decided the importance of each column on the outcome (benign/malicious). From there we decided on certain columns to drop being the timestamp, flow bytes/sec, and flow packets/sec. We then looked to fill in any missing values to make sure all data was present. Due to the dataset being so large, we decided to scale it down into eight separate csv files to test our code faster on smaller datasets. Running visuals such as PCA scatterplot and correlation plots, we noticed that the attacks are centered mostly within the benign data points. We used these visuals to gain a better understanding on how we should further tweak our dataset and run our model to classify the attacks versus the benign instances.

Currently, we chose to use a sequential model from the Keras library. This is a simple linear model with 2 layers of 64 nodes each. We decided to use Keras because it allows us to change the model as needed later down the road. Keras itself is pretty high-level, but due to its integration with tensorflow, we can use low-level features and abilities if we want to

use a more complex model later. Its flexibility allows us to improve/change our neural net model. The neural net training technique was ideal for the way our data was distributed. There was no perfect distinction between the attack and benign points, so the versatility of neural nets will be able to better classify our data.

So far, we've analyzed and cleaned the data, as well as implemented Keras' simple Sequential model, which is a linear stack of layers. If we determine the accuracy of the Sequential model to be unsatisfactory, we will research alternative models that enable a higher level of customization and complexity, ideally a model that allows training data in sequences such as in a Recurrent Neural Network.

Related work

Using Adaptive Alert Classification to Reduce False Positives in Intrusion Detection

This paper appears to be one of the earlier papers to attempt using machine learning to solve the problem of false positive alerts. The model uses a neural network and machine learning to classify attacks by whether or not it is a false positive, and requires an analyst to correctly audit and adjust the program's results to train the data. The paper goes in depth about how crucial it is to provide proper background knowledge about the system to the neural network, to allow it to 'learn' relevant information by classifying certain data before it is fed to the neural network.

Data Mining and Machine Learning: Towards reducing false positives in Intrusion Detection

This paper presents two different, but orthogonal approaches to dealing with the problem of false

positives. The first approach is to cluster many similar alerts together based on a root cause, so that a human reading the alerts has fewer total alerts to sort through. This is a pretty common method of dealing with too many alerts. The paper also uses an additional machine-learning program to either add labels to, or to directly filter the incoming alerts to reduce the number of alerts. The machine learning system essentially attempts to label alerts as 'accurate' or 'false positive' by automated means. These two approaches, clustering and filtering, work in tandem to reduce false positives.

Finding Needles in a Moving Haystack: Prioritizing Alerts with Adversarial Reinforcement Learning

This paper details an interesting approach to prioritizing alerts for intrusion detection systems. The approach involves using Adversarial Reinforcement learning to simulate both a defender and an adversary. This approach seems like it has some potential, because the defender/adversary approach mimics what happens in reality, with an IDS system needing to handle attacks from an adversary, which may be able to change their attack method. In the simulated scenario, the attacker can adjust the likelihood of different attack methods, and the defender must try to accurately prevent false positives in varying conditions.

Alert Fusion for A Computer Host Based Intrusion Detection System

This paper takes a pretty interesting approach to managing the IDS system alerts. The authors have essentially hand-crafted a neural network to determine the alert priority. They base their customized neural network on a "STRIDE/DREAD" threat model, with nodes hand-weighted for certain instances. This model doesn't involve machine learning, but the authors mention that in future, backpropagation can be used

to ‘train’ the network for further improvements. This method is very interesting because a hand-designed setup like this can be understood directly, and tweaked further with ease to adjust the severities of specific types of attacks.

Deep Learning for Prioritizing and Responding to Intrusion Detection Alerts

The paper discusses the results of the Federal Analysis Security Triage Tool (FASTT), which relies on a machine learning approach, employing Deep Neural Network (DNN) classifiers to automatically categorize IDS alerts. The system consists of an IDS and a data pipeline, which retrieves data from the IDS and classifies it using DNN classifiers. The results, recorded in an indexed database, are then presented to the appropriate security analyst through a graphical interface. Classifiers were trained using initial data labeled by security analysts. As security threats evolve, the authors note that the classifiers will improve with additional analyst feedback. The paper suggests future implementations using clustering method may lead to improvements in detection, allowing analysts to identify groupings that were overlooked through the heuristic approach.

A Hybrid High-Order Markov Chain Model for Computer Intrusion Detection

This paper goes over how to use a Markov chain to aid in intrusion detection. A simple Markov chain consists of an input and gives an output along with the probability of said output. In the paper, the inputs used are a series of Unix commands entered. A system utilizing this would have a baseline “signature” for each user in order to model how the user interacts with the command line. By comparing a session with the user’s signature, an IDS will be able to alert about potential anomalies in the commands entered. A similar system could

potentially be expanded to targets such as network traffic. Typical traffic can be recorded and then modeled as a Markov chain using parameters such as typical packet size, connection length/frequency, and source/destination.

Combining Recurrence Quantification Analysis and Adaptive Clustering to Detect DDoS Attacks

This paper discusses using an adaptive algorithm to analyze network traffic and detect a DDoS attack which is more and more important as an increasing number of services rely on the Internet. The base algorithm, Recurrence Quantification Analysis (RQA), allows for mathematical analysis of a non-linear periodic signal to determine data points such as Recurrence Ratio, Determinism, and Entropy. To RQA, they add on what they call Adaptive Clustering (A-Kmeans) which can change the number of clusters used dynamically to better express the attack. This method, dubbed DDoSbyRQA, is an improvement on the traditional method simply analysing network traffic behavior based on a set of attributes. This older method has a major pitfall in that when normal traffic increases, so does the number of false positives.

References

- [1] Pietraszek, Tadeusz. "Using adaptive alert classification to reduce false positives in intrusion detection." International Workshop on Recent Advances in Intrusion Detection. Springer, Berlin, Heidelberg, 2004.
- [2] Pietraszek, Tadeusz, and Axel Tanner. "Data Mining and Machine learning—Towards Reducing False Positives in Intrusion Detection." Information Security Technical Report 10.3 (2005): 169–183. Web.
- [3] Liang, Tong et al. "Finding Needles in a Moving Haystack: Prioritizing Alerts with Adversarial Reinforcement Learning." arXiv.org (2019): n. pag. Web.

- [4] Chuan Feng et al. "Alert Fusion for a Computer Host Based Intrusion Detection System." 14th Annual IEEE International Conference and Workshops on the Engineering of Computer-Based Systems (ECBS'07). IEEE, 2007. 433–440. Web.
- [5] McElwee, Steven, et al. "Deep learning for prioritizing and responding to intrusion detection alerts." MILCOM 2017-2017 IEEE Military Communications Conference (MILCOM). IEEE, 2017.
- [6] Ju, Wen-Hua, and Yehuda Vardi. "A Hybrid High-Order Markov Chain Model for Computer Intrusion Detection." Journal of Computational and Graphical Statistics, vol. 10, no. 2, 2001, pp. 277–295. JSTOR, www.jstor.org/stable/1391012. Accessed 8 Feb. 2020.
- [7] Righi, Marcelo Antonio, and Raul Ceretta Nunes. "Combining Recurrence Quantification Analysis and Adaptive Clustering to Detect DDoS Attacks." The Cyber Defense Review, 2019, pp. 15–30. JSTOR, www.jstor.org/stable/26846118. Accessed 8 Feb. 2020.